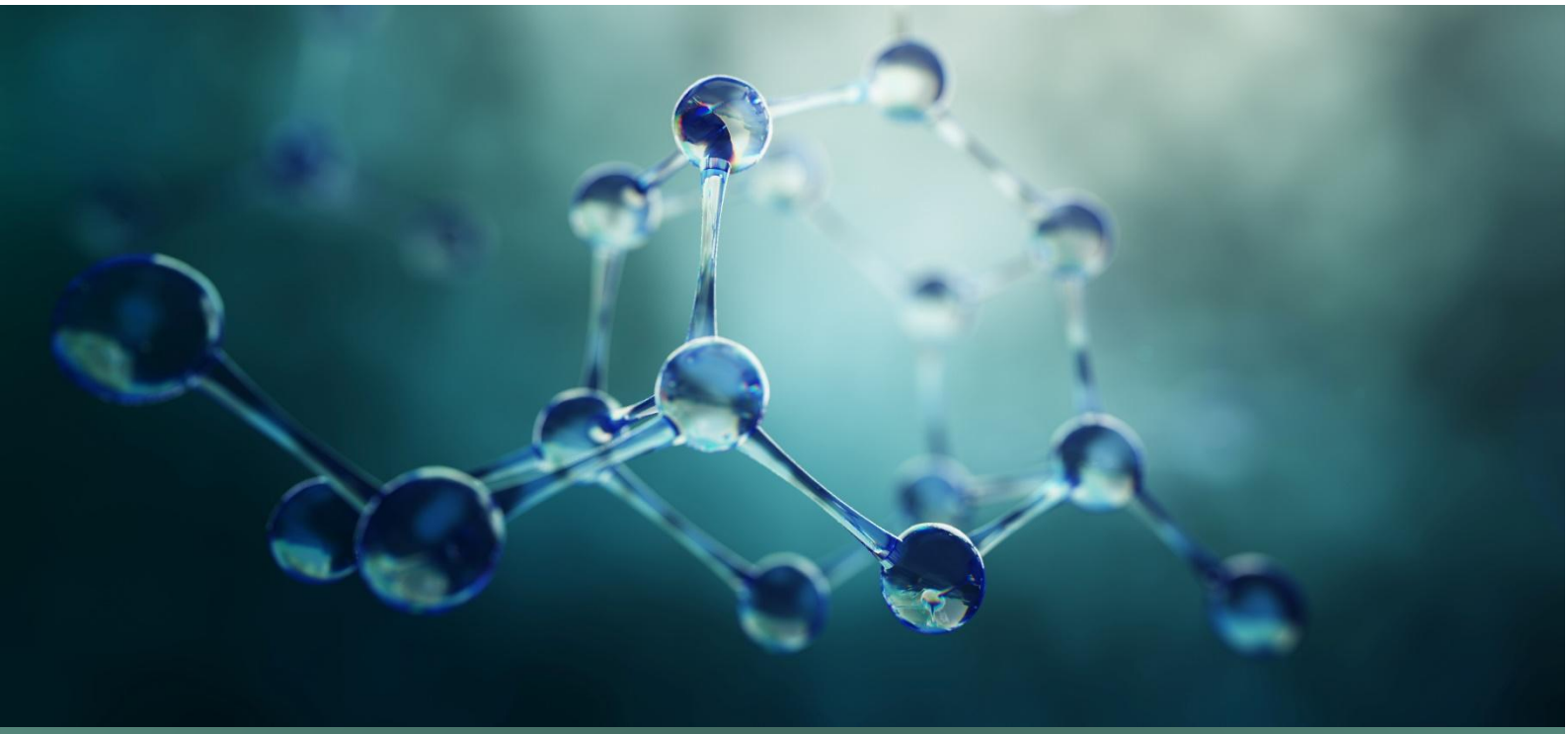
RESULTATEN ONDERZOEK RAPPORT

Intra Air
METEN IS WETEN.



Mould investigation

Location: Westblaak 67,Rotterdam

02-02-2026

Intra Air
Hoofdweg 356
2908LC Capelle aan den Ijssel

Assignment Details

Client	Easy Hotel
Inspection Address	Westblaak 67,Rotterdam
Contact Person	Wim Venema
Property Type	Hotel
Rooms / Measurement Locations	Multiple Rooms
Carried out by	Giancarlo Caruso
Date of Inspection	02-02-2026
Date of Report	13-02-2026

Special Notes

Several of the inspected rooms had been in use shortly prior to the inspection. A number of other rooms were not in use due to mould growth.

Investigation Findings

- An elevated concentration of mould spores was identified in the inspected rooms.
- Visible mould growth was observed in several rooms.
- Significant mould growth is suspected behind the thick renovation fleece/glass fibre wallpaper.
- The ventilation system is not functioning properly in several rooms, resulting in elevated CO_2 concentrations.
- The mechanical ventilation unit on the roof shows visible mould contamination.
- Condensation was observed in multiple rooms, consistent with insufficient ventilation and inadequate air circulation.
- Leakage was identified around several shower enclosures.
- Water ingress from the roof was identified on the upper floor.

Below is a detailed description of the investigation, the results, and our recommendations.

Dear Mr Venema,

On 2 February 2026, we carried out a mould investigation on your behalf. In this report, we present our findings, conclusions, and recommendations. The assessment represents a snapshot of the general condition of the indoor environment at the time of the inspection.

During the investigation, we conducted a visual technical inspection, collected samples, and performed various measurements in several rooms.

RESULTS OF THE INVESTIGATION AND FINDINGS

I. Average Results of the Air Quality Assessment:

The standard air quality assessment includes measurements of temperature, relative humidity (RH), carbon dioxide (CO_2), and particulate matter (PM 2.5). Deviations from the normal values of these four parameters may contribute to various health complaints.

For this assessment, ten measurement locations were selected, namely rooms numbered: 105, 111, 203, 309, 406, 415, 509, 601, 602, and 611.

Room number	Temp (18-22°C	RH (40-60%)	CO_2 (400-850 PPM)	PM (> P.M. 2.5)
105	28 °C	36%	730 PPM	2700
111	21 °C	66%	675 PPM	2300
203	28 °C	37%	750 PPM	2500
309	28 °C	67%	1400 PPM	2900
406	24 °C	40%	1150 PPM	2800
415	27 °C	57%	1360 PPM	2900
509	22 °C	65%	1000 PPM	3400
601	28 °C	59%	1050 PPM	3700
602	25 °C	66%	980 PPM	3300
611	21 °C	70%	760 PPM	7500

The results of the standard air quality assessment demonstrate that elevated levels of temperature, relative humidity, carbon dioxide (CO_2), and particulate matter were identified in multiple rooms.

CO2

Carbon dioxide (CO_2) is a crucial indicator of indoor air quality. When CO_2 levels exceed 1,000 ppm, various health effects may occur, including respiratory discomfort, headaches, reduced concentration, and a general feeling of malaise.In general, CO_2 concentrations between 400 and 850 ppm are considered acceptable, while values above this range require additional attention to ensure a healthy indoor environment.

RH

An indoor environment with unsuitable humidity levels may lead to health issues, particularly for individuals with asthma or increased sensitivity. The ideal relative humidity generally ranges between 40% and 60%; however, for people with asthma it is often recommended to maintain levels between 40% and 50%.

Excessively humid conditions also promote mould growth.

PM

Particulate matter is a collective term for airborne pollutant particles that are small enough to be inhaled and may cause damage to the lungs. These particles can originate from various sources and vary in composition.

Exposure to particulate matter is harmful to health and may also indicate the presence of mould spores. We measured particulate matter using a particle indicator to gain insight into the air quality within the property.

II. Additional Air Quality Measurements

A Additional air quality measurements were carried out to assess oxygen levels and the presence of formaldehyde (CH_2O) and/or volatile organic compounds (VOCs). Exceedance of the guideline values for these three parameters may also result in health complaints.

LOCATION	Oxygen	CH_2O	VOC
Hotel	20.9%	None	None

The additional air quality measurements were carried out using a personal gas detector worn during the inspection for personal safety purposes. No points of concern were identified during these measurements. However, it should be noted that this did not constitute an in-depth investigation, as the primary focus of the assessment was mould-related issues.

III. Microbiological Air Investigation Results

Air samples were collected using an air sampler, drawing moulds and yeasts onto OGEY agar plates at multiple locations within the property. In addition, a reference sample was taken from the outdoor air.

Surface contact samples were collected using RODAC contact plates, whereby moulds and yeasts were sampled from various surfaces in the rooms by briefly pressing the agar surface against the substrate.

The results of this investigation indicate a very high concentration of mould spores in several hotel rooms.

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 105	Air	Mould	Aspergillus species	170 cfu/m³ *
			Penicillium species	230 cfu/m³ *
			Cladosporium species	180 cfu/m³ *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 111	Air	Mould	Aspergillus species	320 cfu/m³ *
			Acremonium species	100 cfu/m³ *
			Penicillium species	210 cfu/m³ *
			Cladosporium species	130 cfu/m³ *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 203	Contact	Mould	Aspergillus species	160 cfu/plate *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 309	Air	Mould	Mucor species	160 cfu/m³ *
			Aspergillus species	140 cfu/m³ *
			Penicillium species	170 cfu/m³ *
			Cladosporium species	110 cfu/m³ *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 406	Air	Mould	Aspergillus species	290 cfu/m³ *
			Penicillium species	210 cfu/m³ *
			Cladosporium species	190 cfu/m³ *
			Rhizopus species	20 cfu/m³ *

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Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 415	Air	Mould	Aspergillus species	240 cfu/m³ *
			Penicillium species	210 cfu/m³ *
			Cladosporium species	180 cfu/m³ *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 509	Contact	Mould	Aspergillus species	Overgrowth *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 601	Air	Mould	Aspergillus species	250 cfu/m³ *
			Penicillium species	290 cfu/m³ *
			Cladosporium species	160 cfu/m³ *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 602	Contact	Mould	Aspergillus species	Overgrowth *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Room 611	Air	Mould	Penicillium species	>500 cfu/m³ *
			Aspergillus species	>500 cfu/m³ *
			Rhizopus species	110 cfu/m³ *

Sample location	Type	Species	Identified Mould Species	Total Quantity
Outdoor Reference Sample	Air	Mould	Aspergillus species	
			Penicillium species	
			Cladosporium species	
				116 cfu/m³ *

IV. Conclusion

We conducted an extensive investigation within the hotel, during which multiple rooms were inspected and sampled. Depending on the conditions in each room, both air samples and surface contact samples were collected. In addition, carbon dioxide (CO_2), relative humidity, temperature, and particulate matter were measured in every room. Moisture measurements of building surfaces were also performed to assess the presence of moisture-related issues.

The investigation revealed significant problems throughout the hotel. Ten rooms were examined, and mould was identified in all ten rooms. This included both airborne mould spores and numerous visible areas of active mould growth. The identified mould species were largely similar across the rooms and included Aspergillus species, which are known to be associated with various health complaints.

A key issue identified during the investigation is that many mould-affected areas are not immediately visible. Mould was often only detected after targeted inspection. In several rooms, mould growth was identified behind beds (including room 203) and behind thick renovation fleece wallpaper, which was observed in multiple rooms.

Furthermore, several rooms showed signs of minor leakage originating from the shower area, including built-in shower cabins and bathtubs. In addition, the mechanical ventilation system was found to be functioning inadequately. In multiple rooms, CO_2 concentrations were extremely high. Even after rooms had been unoccupied for more than one hour, CO_2 levels remained elevated. This is indicative of insufficient ventilation capacity, poorly adjusted mechanical ventilation, and contamination of the ventilation unit. It is also likely that the ventilation ducts are in need of thorough cleaning.

In particular, air extraction appears to be insufficient. Air is extracted only in the bathroom, while supply air is introduced above the bed. This configuration results in poor air circulation within the room. Additionally, all rooms are equipped with air conditioning units that blow air directly into the supplied ventilation air, causing airflow interference. Smoke testing carried out during the inspection clearly demonstrated inadequate air circulation within the rooms.

In several rooms, ventilation grilles were partially blocked with paper, causing further obstruction of airflow. Moreover, temperatures in multiple rooms were set as high as 28°C, resulting in warm and stuffy conditions. Elevated temperatures, particularly when combined with increased humidity and insufficient ventilation, create favourable conditions for mould growth and significantly increase the risk of mould development.

Inspection of the rooftop mechanical ventilation system revealed mould growth inside the air handling unit. As supply air is distributed from this unit to the rooms, there is a high likelihood that air entering the rooms already contains elevated concentrations of mould spores. This facilitates the spread of mould throughout the entire building. When combined with inadequate room usage conditions, such as insufficient ventilation and high moisture production, the development and persistence of mould growth within the hotel is considered highly plausible.

Proposed Measure

Ten rooms were investigated, in nearly all of which visible mould growth was identified. In some rooms, the extent of mould growth was significantly more severe. Room 611, which was the primary reason for initiating this investigation, showed extensive mould contamination. The mould became clearly visible only after a staff member had removed part of the wallpaper shortly before the inspection, causing a large amount of mould-contaminated material to detach.

In other rooms, mould growth was identified behind beds and behind wallpaper, becoming visible only after removal of the wall covering. This leads to the assumption that additional rooms may be affected by similar hidden mould problems that have not yet been identified.

It is therefore strongly recommended to carry out targeted inspections of all rooms, focusing on vulnerable areas. Wallpaper should be locally removed at critical points to assess whether mould growth is present behind it. Early detection is essential, as untreated mould growth may continue to spread behind wall finishes and in poorly ventilated areas such as behind beds. Although these inspections may involve significant intervention, timely action will help ensure that rooms remain hygienic and suitable for hotel guests.

It is advised to carry out these inspections and remediation activities on a floor-by-floor basis. This phased approach allows the hotel to remain operational while inspections, cleaning, and repairs are conducted during quieter periods.

The mechanical ventilation system requires thorough investigation and cleaning. This includes both the air handling unit located on the roof and the associated ductwork. A comprehensive assessment is essential, as there is no natural ventilation within the building and this system is responsible for supplying and extracting air throughout the hotel. If the system is not functioning correctly, as currently appears to be the case, the basic indoor climate within the hotel cannot be adequately maintained. Even in unoccupied rooms, contaminated air may currently be distributed throughout the building. It is therefore strongly recommended to inspect, measure, and clean the ventilation system thoroughly. Improving the base air treatment will significantly enhance indoor air quality throughout the hotel.

All identified mould-affected areas should be professionally removed using appropriate methods and materials. Following remediation, it is recommended to apply a mould-resistant coating to the walls before reinstalling wallpaper. In areas where issues with the external façade are known, it may be preferable not to reapply wallpaper and instead finish the walls with paint only. Wallpaper can trap moisture, increasing the risk of recurrent mould growth.

It is also recommended to measure moisture levels around all shower enclosures to determine whether leakage from the shower areas is present. Based on these measurements, a targeted action plan can be developed. In many cases, the issue can be resolved by replacing sealant; however, if such issues are not addressed in a timely manner, secondary damage may occur, such as floor deformation or detachment of wallpaper, both of which increase the risk of mould growth. Systematic inspection and documentation of moisture conditions in the bathrooms can significantly reduce repair costs when addressed at an early stage.

Summary of Recommendations

- Conduct targeted inspections in all rooms by removing wallpaper at vulnerable locations to identify hidden mould growth.
- Carry out inspections and remediation work on a floor-by-floor basis to allow the hotel to remain operational during quieter periods.
- Thoroughly inspect, measure, and clean the complete mechanical ventilation system, including the rooftop air handling unit and all ductworks.
- Improve ventilation performance to ensure adequate air circulation and prevent the spread of mould spores throughout the building.
- Professionally remove all identified mould-affected materials using appropriate remediation methods.

- Apply a mould-resistant coating to treated walls prior to reinstalling wallpaper.
- Avoid the use of wallpaper in areas with known façade-related moisture issues; apply paint finishes instead where appropriate.
- Measure moisture levels around all shower enclosures to identify potential leakage.
- Repair identified leaks promptly, including replacement of sealants where required, to prevent secondary damage and mould growth.

Our Support.

Should you wish, we can assist you with further investigations and/or carry out the recommended remediation works. If required, we would be pleased to prepare a tailored quotation for both the inspection and cleaning of the complete ventilation system, as well as for the investigation and professional remediation of mould contamination in the hotel rooms.

Please let us know if you would like us to proceed, and we will provide a detailed proposal and cost estimate accordingly.

In preparing this report, we have endeavoured to be as thorough as possible.

Should you have any questions regarding this report or the investigation carried out, please feel free to contact us on weekdays between 9:00 and 17:00 by telephone on +31 6 1873 8897 or by email at info@intra-air.nl

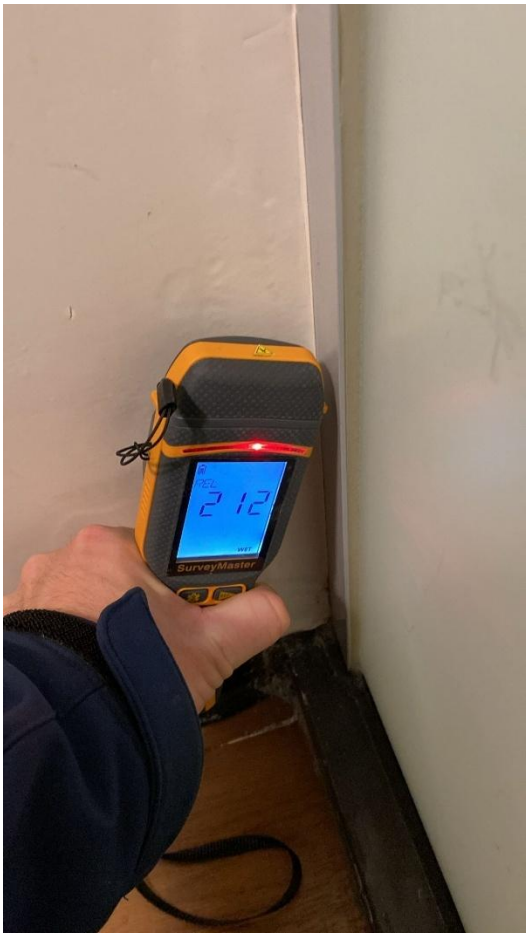
Appendix 1: Guideline Values for Biological Agents

Tabel NVvA/VLA richtwaarden voor totaal koloniegetal per groep micro-organismen voor ruimteluchtmetingen					
De toetsing van de analyseresultaten (kolonievormende eenheden (KVE) per m³ vindt plaats op basis van onderstaande richtwaarden, de buitenluchtconcentraties en de omstandigheden ten tijde van de metingen. Deze kunnen niet los van elkaar beoordeeld worden. De interpretatie van de meetwaardes die getoetst worden aan onderstaande tabel, ligt genuanceerder dan het toetsen aan de absolute waardes. Het doel is om te kijken of er afwijkende waardes gevonden worden (afwijkend van de normaal voorkomende concentraties), welke aanleiding geven tot het nemen van maatregelen of een nader onderzoek. Voor wat betreft het ziekteverwekkend vermogen van de micro-organismen dient gebruik te worden gemaakt van de Europese Richtlijn 2000/54/EG en de daarvan afgeleide Nederlandse wet- en regelgeving.					
Het totaal koloniegetal (aantal KVE/m³) geeft een indicatie over de mogelijkheid voor groei van micro-organismen in de luchtomgeving en bevat alle aangetroffen micro-organismen (som van alle geïdentificeerde en niet geïdentificeerde micro-organismen).					
Beoordeling totaal koloniegetal				**Aantal KVE/m³**	
Geen actie:				< 2500	
Actie vereist:				≥ 2500	
Som aantal micro-organismen uit: Groep A; Groep B; Groep C; overige bacteriën, overige gisten en overige schimmels					
Beoordeling per groep van bacteriën, gisten en schimmels					
Groep A		**Groep B**		**Groep C**	
Micro-organismen uit deze groep kunnen irritatie/klachten veroorzaken bij aantallen boven vermelde richtwaarde.		Irritatie/klachten mogelijk bij lagere aantallen dan groep A.		Zelden voorkomend in kantoren. Indien aanwezig nader onderzoek naar bron.	
Beoordeling bacteriën					
	Aantal per groep KVE/m³		**Aantal per groep KVE/m³**		**Aantal per soort KVE/m³**
Geen actie: Actie vereist:	< 1000 ≥ 1000	Geen actie: Actie vereist:	< 100 ≥ 100	Geen actie: Actie vereist:	< 10 ≥ 10
Bacillus soorten[1] Micrococcus/Kocuria soorten[1] Staphylococen (niet aureus)[1]		Enterobacteriaceae[2] Staphylococcus aureus		Microbispora soorten Micromonospora soorten Thermofiele actinomyceten	
Beoordeling gisten en schimmels					
	Aantal per groep KVE/m³		**Aantal per groep KVE/m³**		**Aantal per soort KVE/m³**
Geen actie: Actie vereist:	< 500 ≥ 500	Geen actie: Actie vereist:	< 100 ≥ 100	Geen actie: Actie vereist:	< 10 ≥ 10
Cladosporium soorten[4] Penicillium (incl. Talaromyces soorten)[4]		Alternaria soorten[4] Aspergillus (overig, incl. fumigatus)[4] Aureobasidium soorten Candida soorten Mucor soorten Phoma soorten Rhizopus soorten Rhodutorula soorten		Acremonium soorten[3] Aspergillus flavus Aspergillus niger Chaetomium soorten[3] Fusarium soorten Scopulariopsis soorten[3] Stachybotrys soorten[3]	
[1] - Indicator organisme voor de mate van schoonmaken van de werkruimten. [2] - Indicator organisme voor persoonlijke hygiëne. [3] - Indicator organisme voor vochtproblemen of waterschade. [4] - Komt veelvuldig voor in de buitenlucht en bij open ramen en deuren kan de binnenluchtconcentratie beïnvloed worden.					
Om de vuistregels toe te kunnen passen dient voor de metingen bij voorkeur gebruik gemaakt te worden van een gevalideerd meetprotocol conform de NEN-EN13098.					

Appendix 2: Photographs.

Room 105

Room 203



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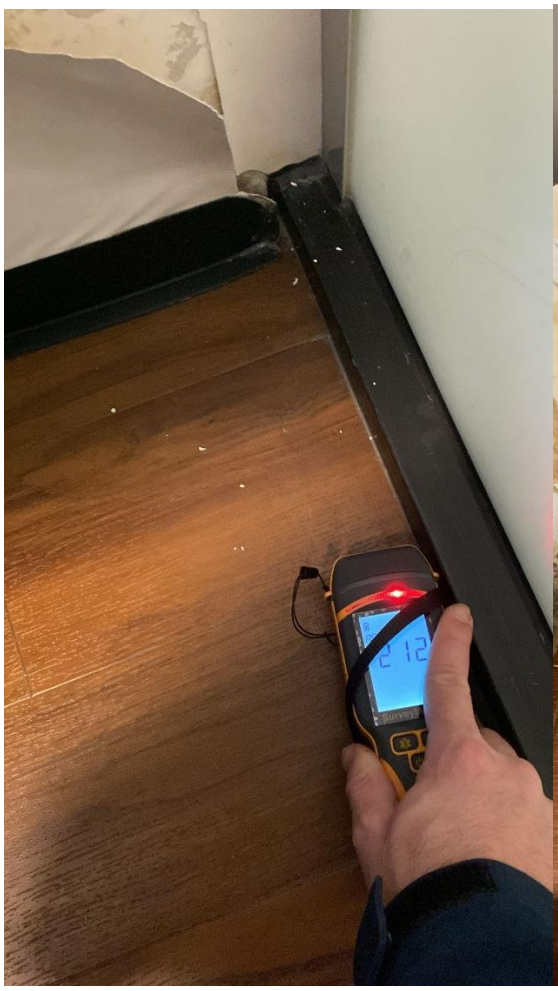
Room 406 smoke test for airflow check

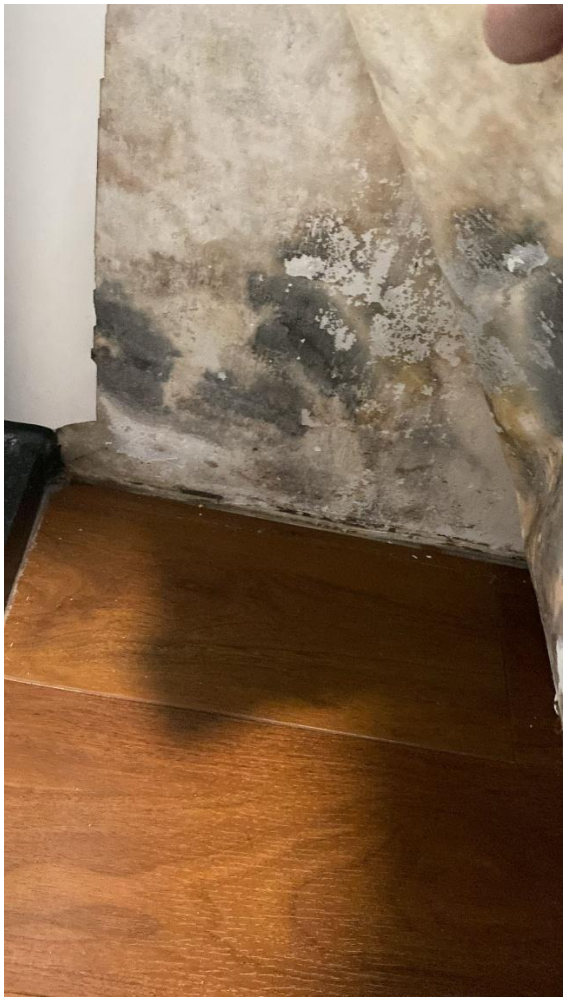
Room 416

Room 509

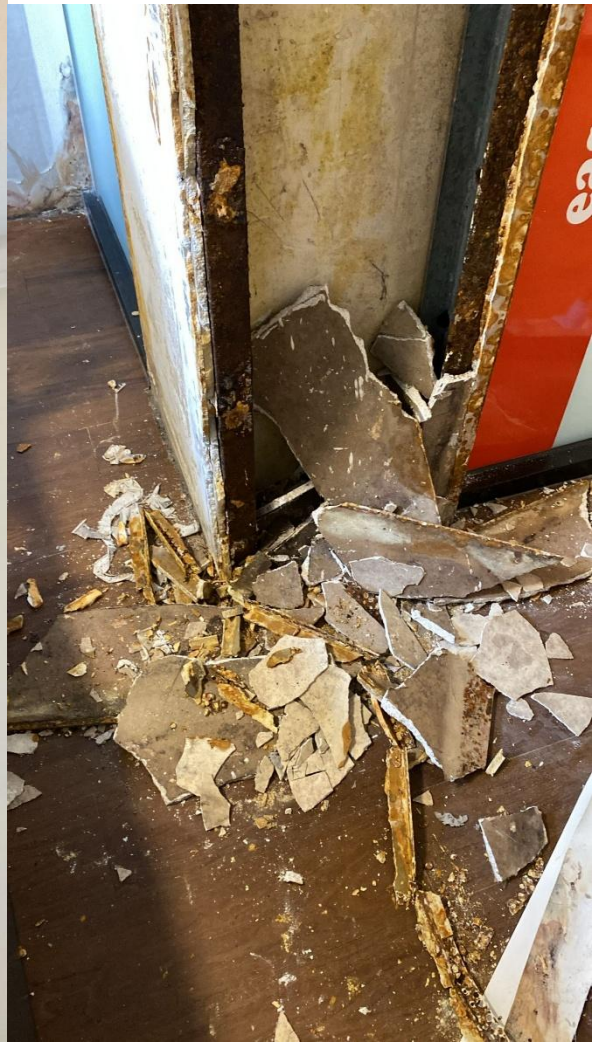
Room 601

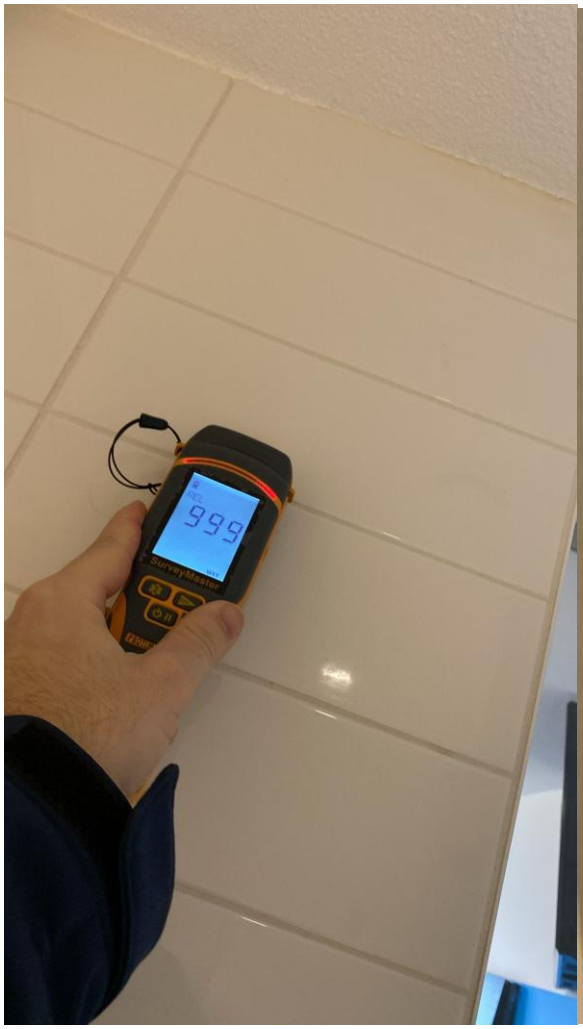


Room 602



Room 611

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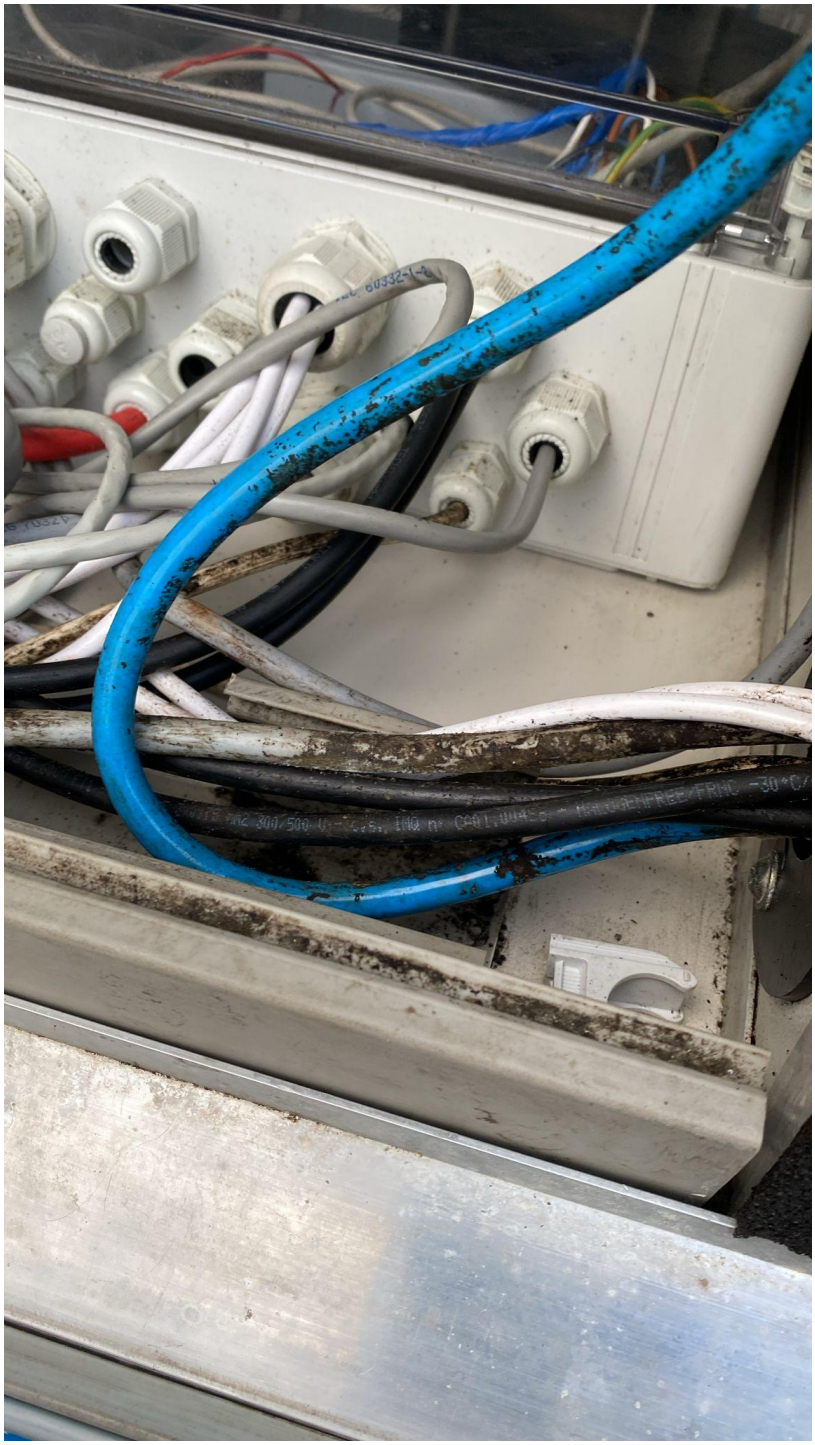
Ventilation Unit (Roof)